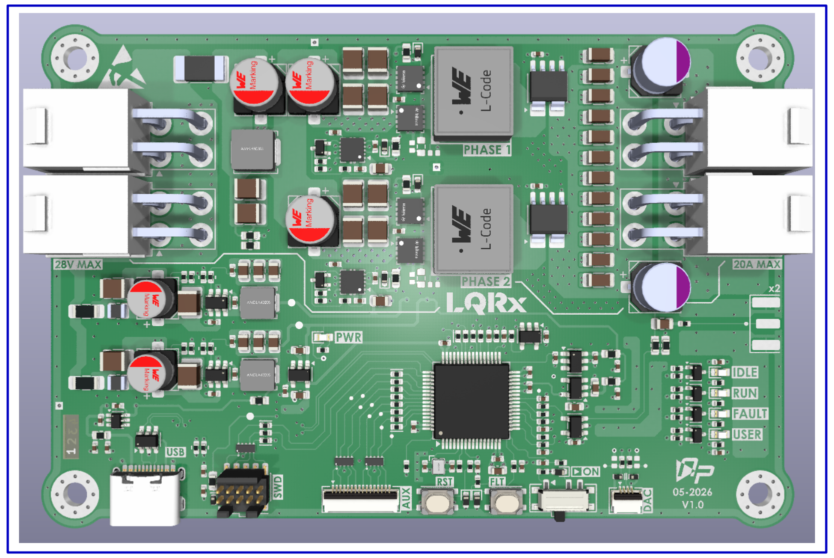
2-Phase Interleaved Sync-Buck Converter

Revision 1.0

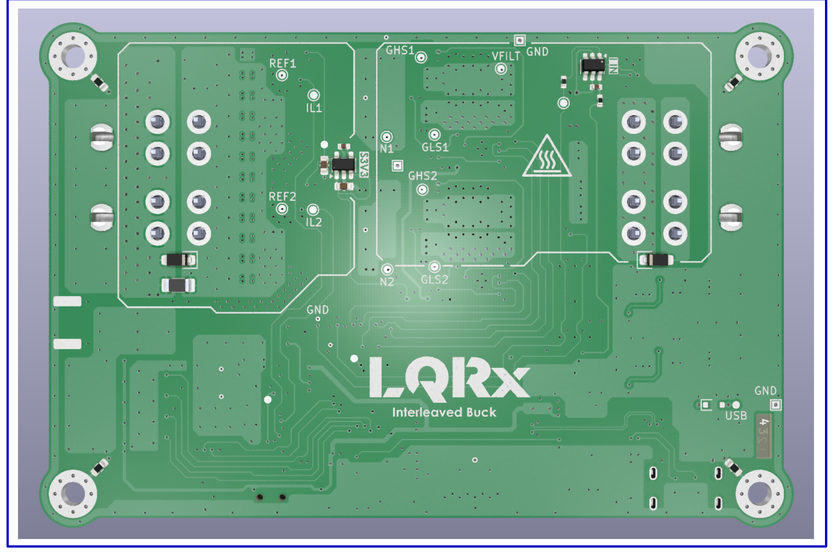
Digitally Controlled Power Conversion with STM32G474

Table of Contents

TOP VIEW



BOTTOM VIEW



Legend

DESIGN NOTE: Text for information design.	DEBUG NOTE: Text for Debug notes.	DESIGN NOTE: Text for design guideline	DESIGN NOTE: Text for critical design	LAYOUT NOTE: Text for layout guideline	LAYOUT NOTE: Text for Critical layout

General Note

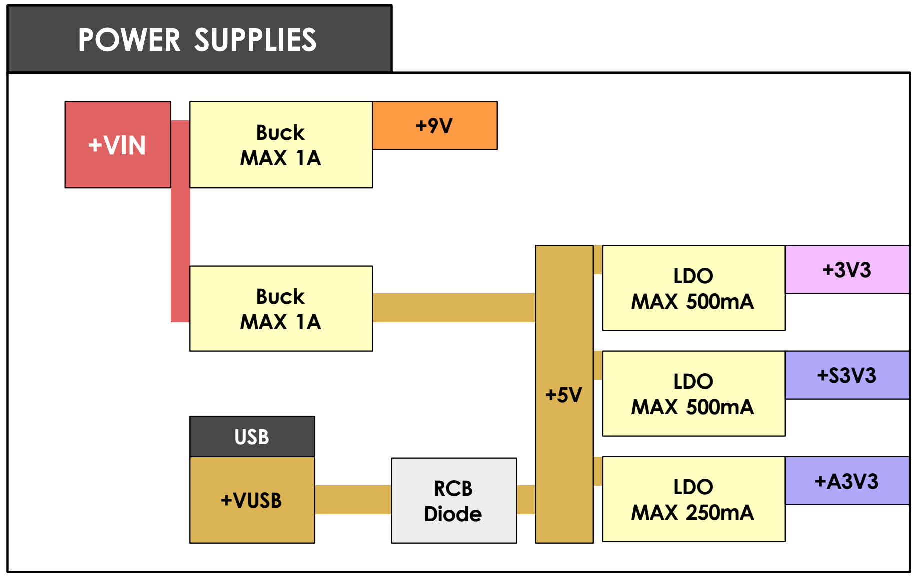
Warning Note

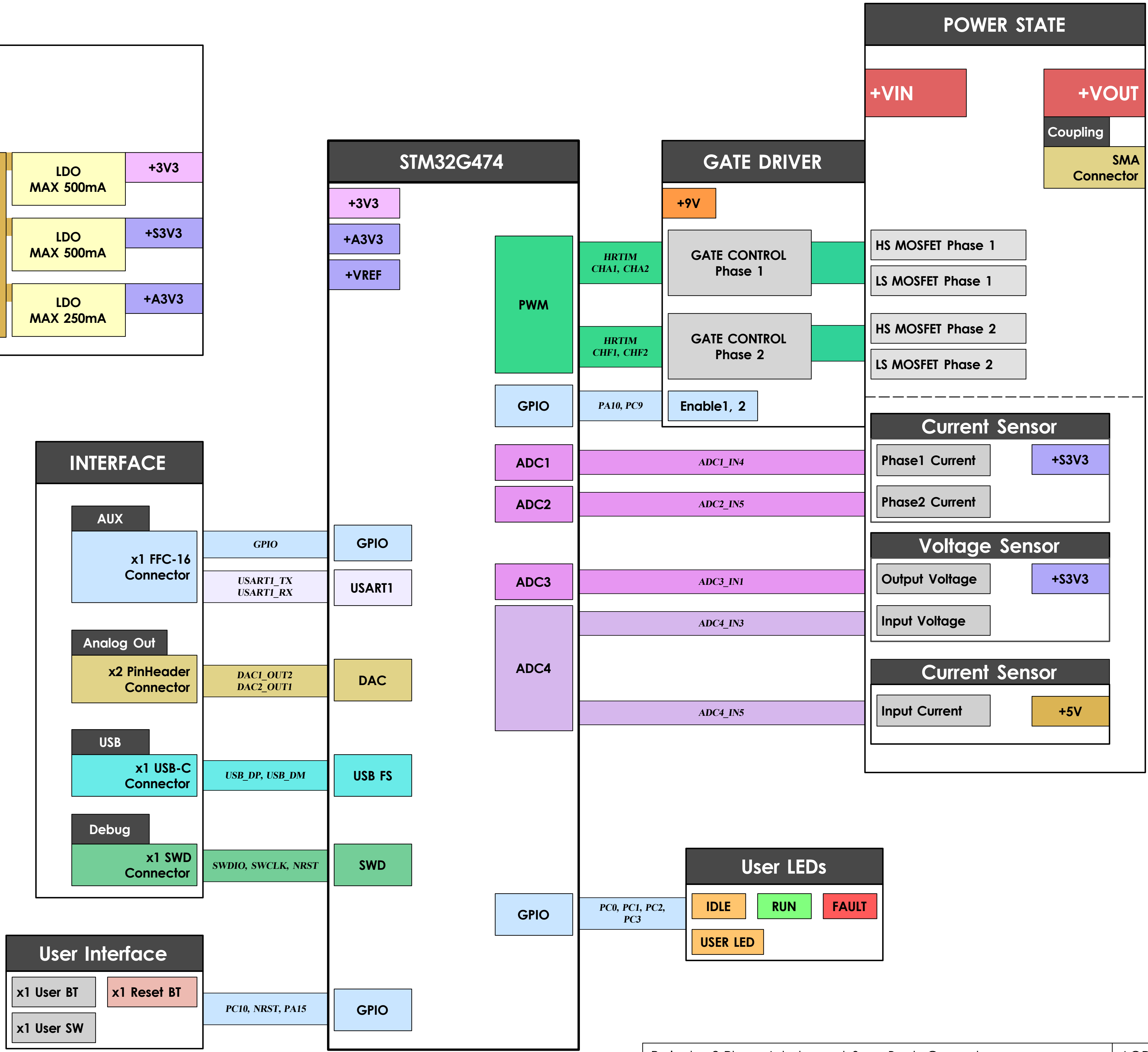
Text to be added to silkscreen.

LQRx

Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: **Project Overview Cover Page**	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: **1.0** Date: 03/06/2026
	Sheet: /		Size **A3** Sheet: **A** of **9**
KiCad E.D.A. 10.0.3	File: 2Phase_SynchBuck.kicad_sch		

[B] Block Diagram



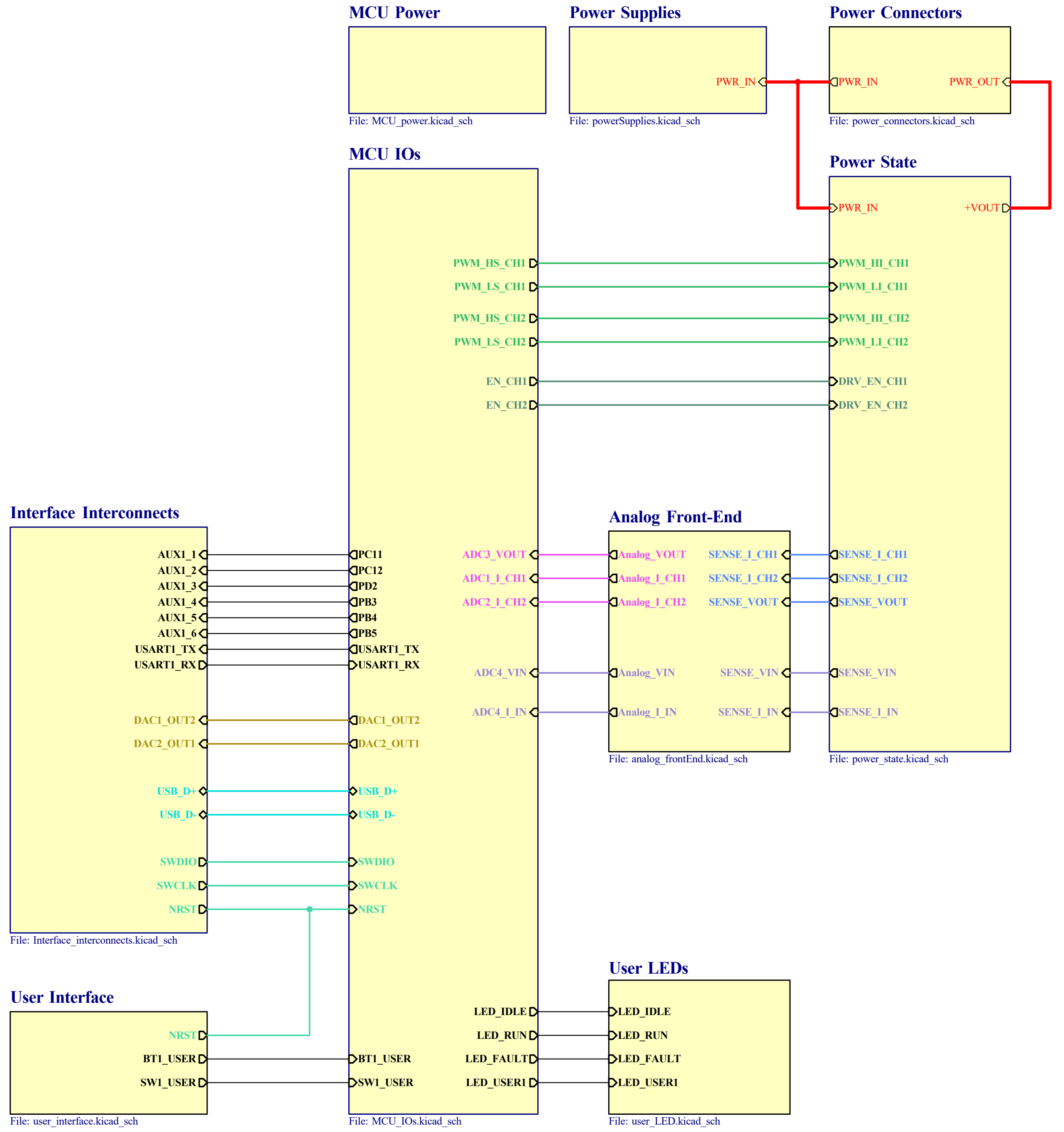


Maximun Rating		MIN	MAX
V_{IN}	Input Voltage	0V	28V
T_{stg}	Storage Temperature	0°C	100°C

Target Specifications		MIN	NOM	MAX
V_{IN}	Input Voltage	15V	24V	28V
V_{OUT}	Output Voltage		12V	
I_{OUT}	Output Current		10A	20A
f_{sw}	Switching Freqency		200kHz	

Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: **Block Diagram (Top Level View)**	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.0 · Date: 03/06/2026
	Sheet: /Block Diagram/		Size A3 · Sheet: B of 9
KiCad E.D.A. 10.0.3	File: Block_diagram.kicad_sch		

[C] Top Hierarchical View



Project: 2-Phase Interleaved Sync-Buck Converter		LQRx		
Title: **Top Hierarchical View (Top Level View)**	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:	
	Reviewed by:	Date:	Rev: **1.0**	Date: 03/06/2026
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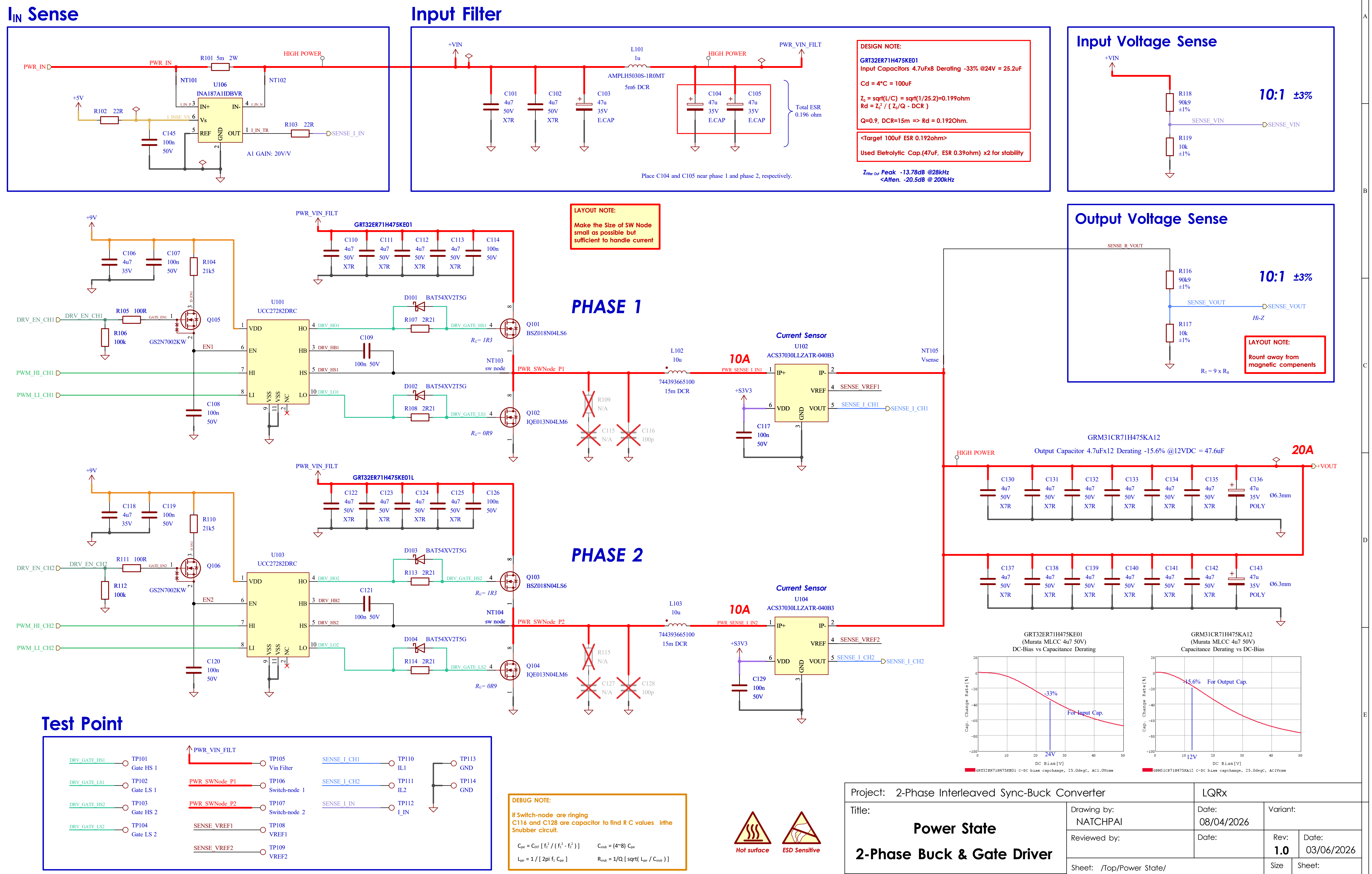
KiCad E.D.A. 10.0.3	File: Top_hierarchical.kicad_sch			

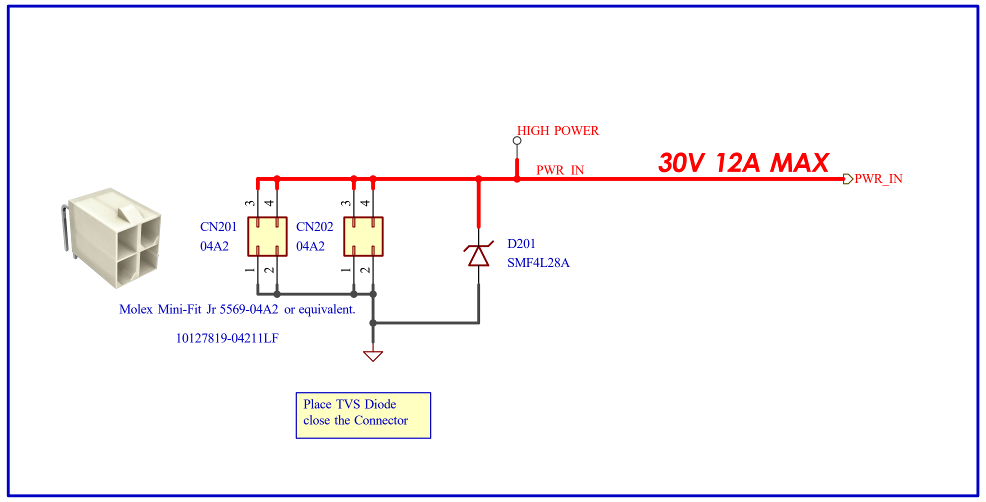
[1] Power State



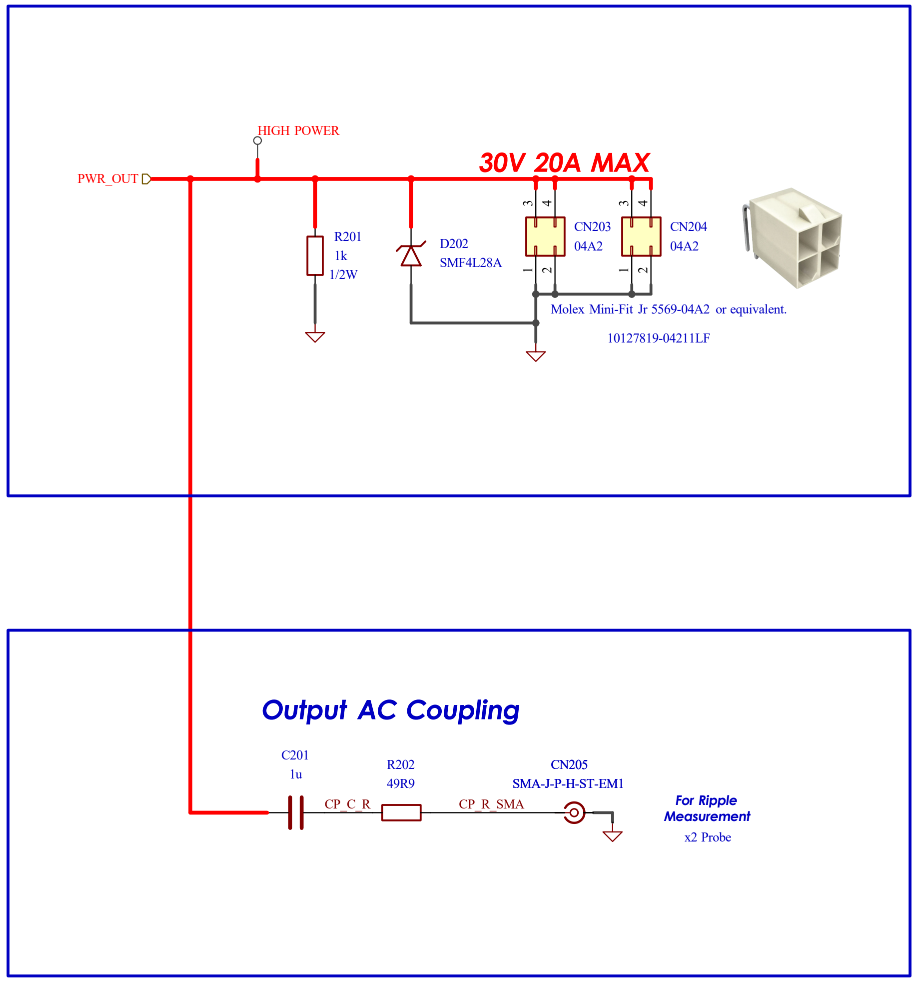
Project:	2-Phase Interleaved Sync-Buck Converter — LQRx
Title:	Power State / 2-Phase Buck & Gate Driver
Drawing by:	NATCHPAI — Date: 08/04/2026
Reviewed by:	Date:
Variant:	
Rev:	1.0 — Date: 03/06/2026
Sheet:	/Top/Power State/
File:	power_state.kicad_sch
Size	A3
Sheet:	1 of 9
KiCad E.D.A. 10.0.3	

[2] Power Connectors

Input Connector



Output Connector



Project: 2-Phase Interleaved Sync-Buck Converter
LQRx

Title: **Power Connectors**

Drawing by: NATCHPAI
Date: 08/04/2026
Variant:
Reviewed by:
Date:
Rev: **1.0**
Date: 03/06/2026
Sheet: /Top/Power Connectors/
File: power_connectors.kicad_sch
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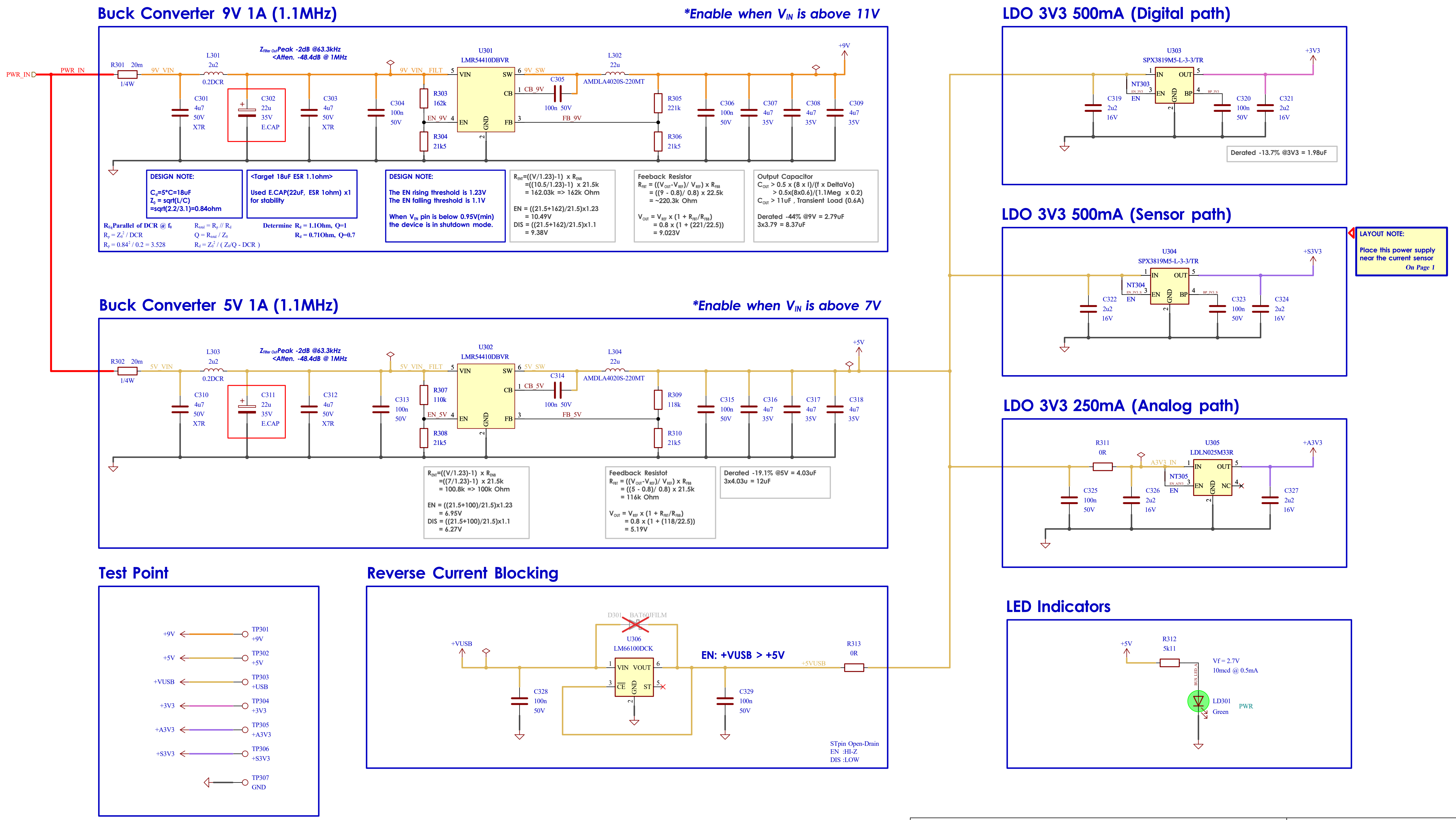
KiCad E.D.A. 10.0.3

[3] Power Supplies



Project: 2-Phase Interleaved Sync-Buck Converter		LQRx		
Title: Power Supplies Auxiliary 9V 5V 3V3	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:	
	Reviewed by:	Date:	Rev: 1.0	Date: 03/06/2026
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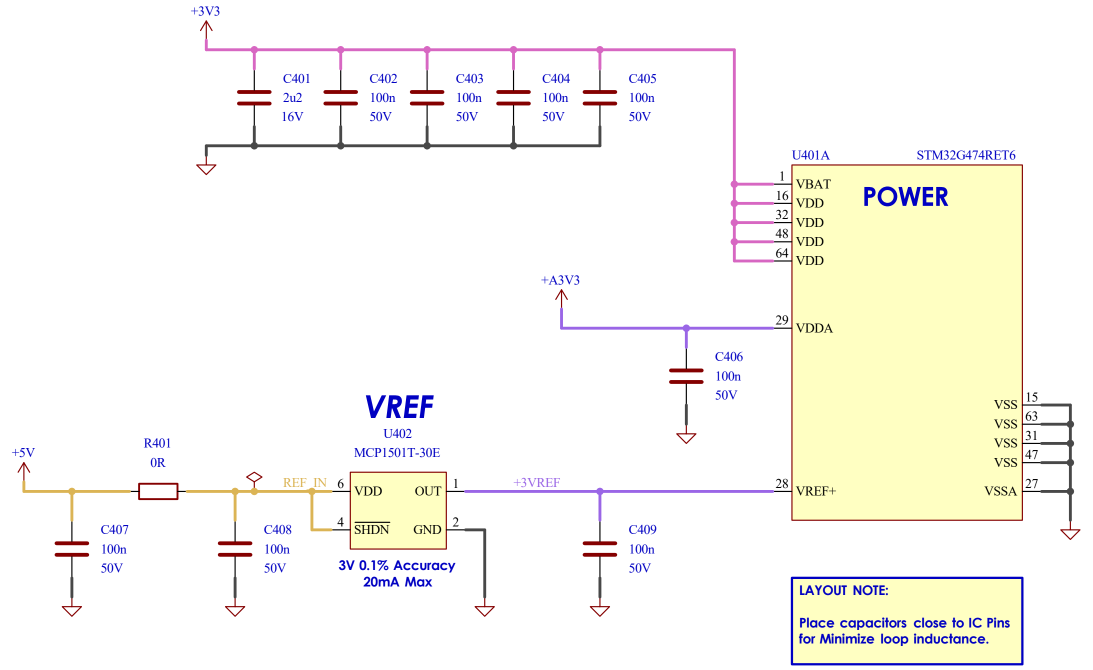
KiCad E.D.A. 10.0.3	File: powerSupplies.kicad_sch			

[4] Microcontroller Power



Project: 2-Phase Interleaved Sync-Buck Converter		LQRx		
Title: **Microcontroller Power** **STM32G474Rx**	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:	
	Reviewed by:	Date:	Rev: **1.0**	Date: 03/06/2026
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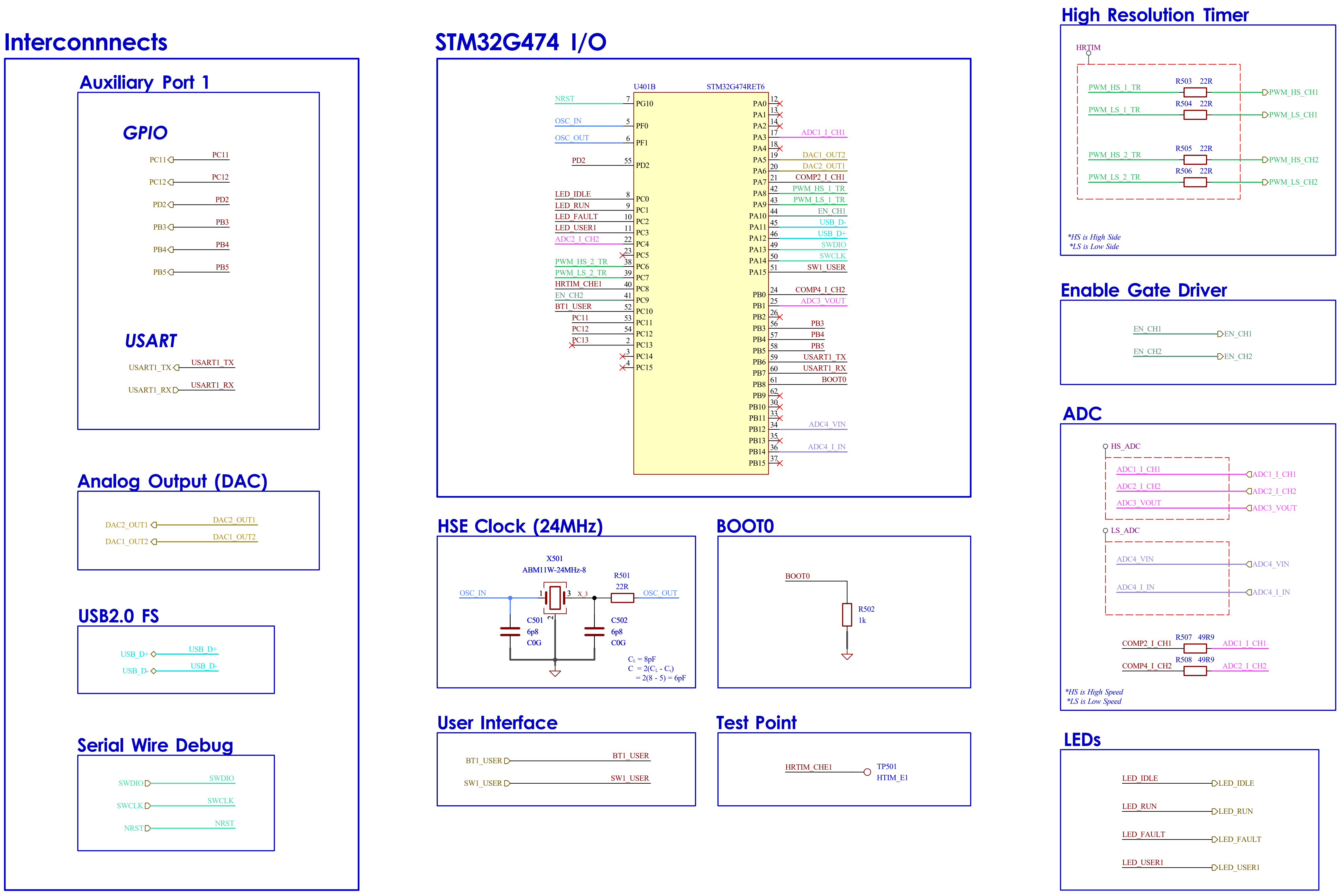
[5] Microcontroller IOs



Project: 2-Phase Interleaved Sync-Buck Converter		LQRx		
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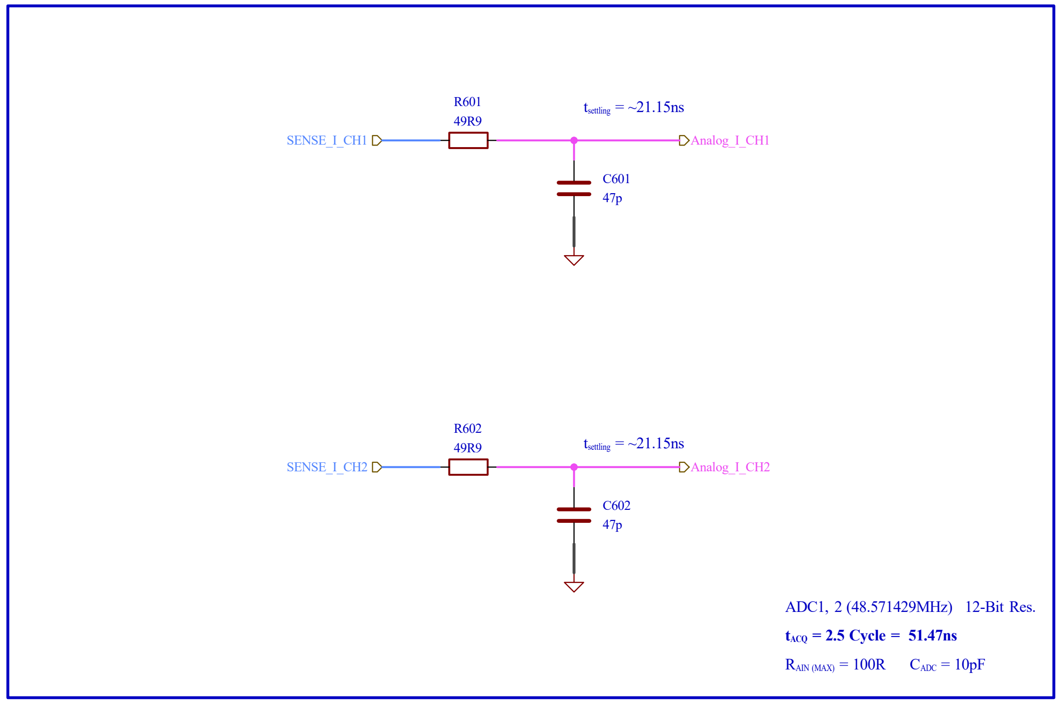
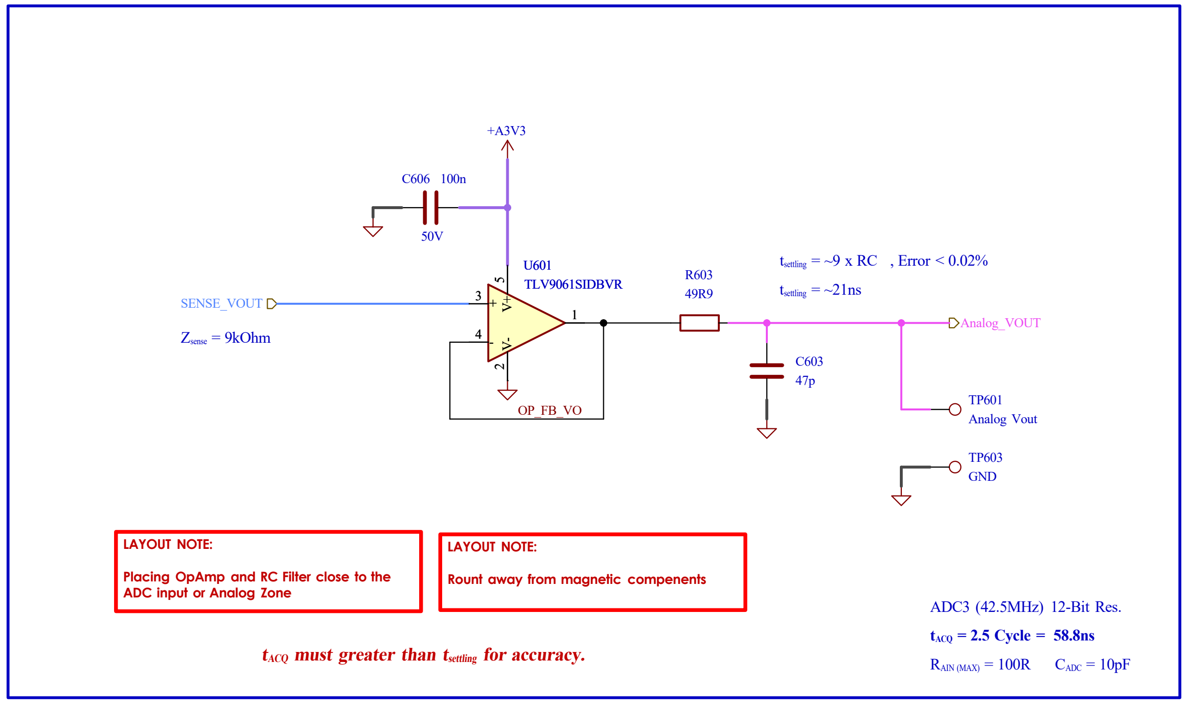
[6] Analog Front-End

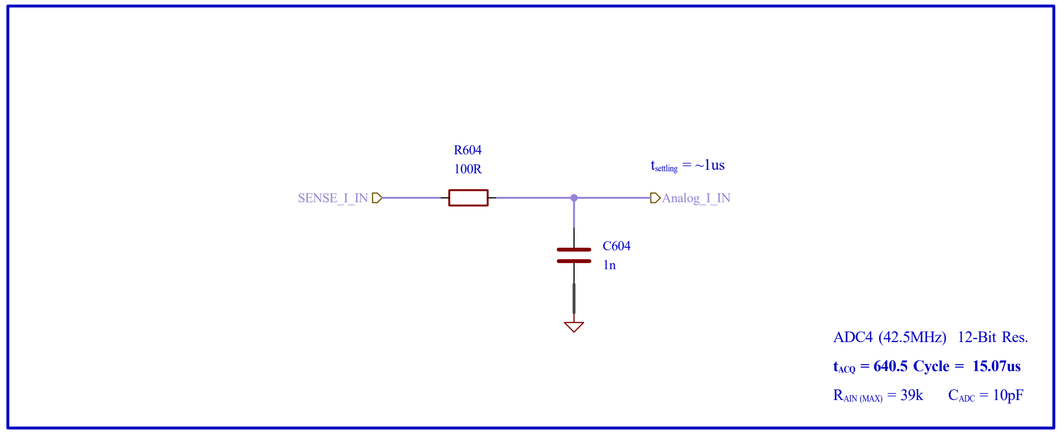
Current Path (High Speed)



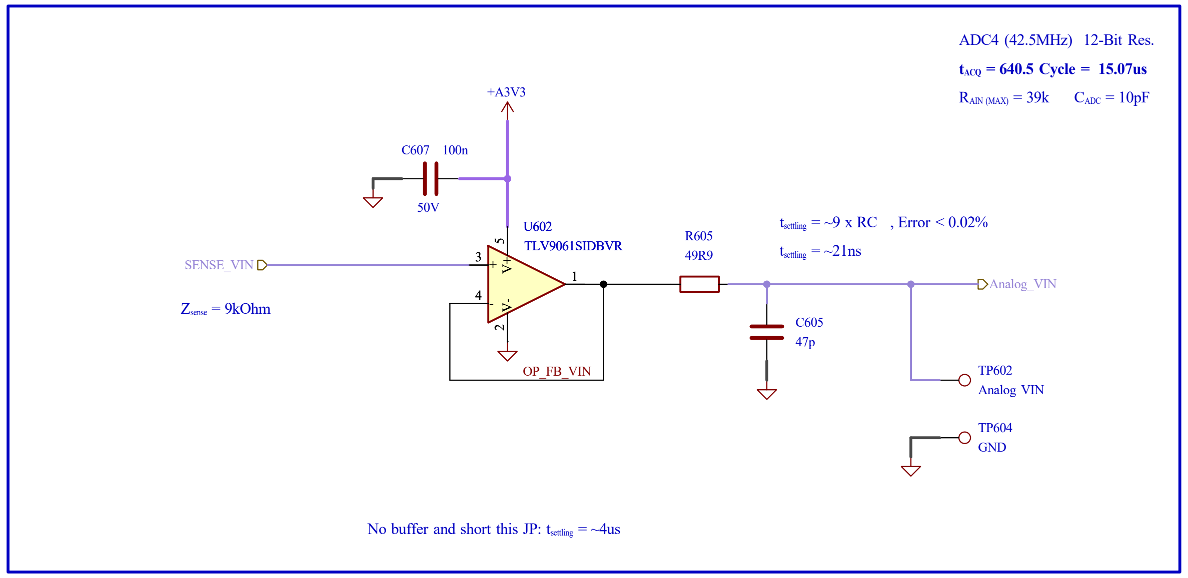
Output Voltage Path (High Speed)



Input Current Path (Low Speed)



Input Voltage Path (Low Speed)



Project: 2-Phase Interleaved Sync-Buck Converter		LQRx		
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	Reviewed by:	Date:	Rev: **1.0**	Date: 03/06/2026
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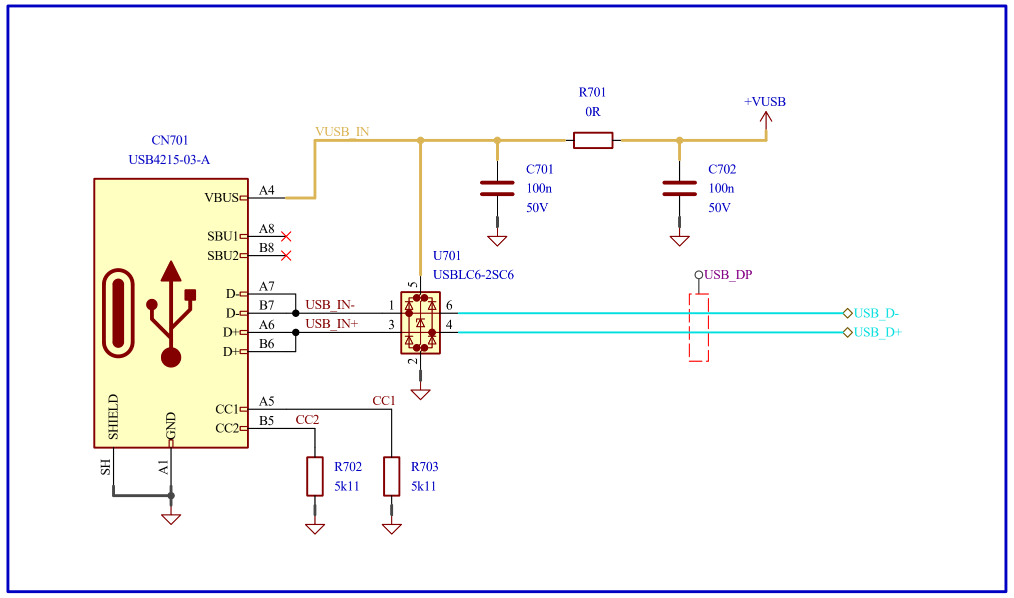
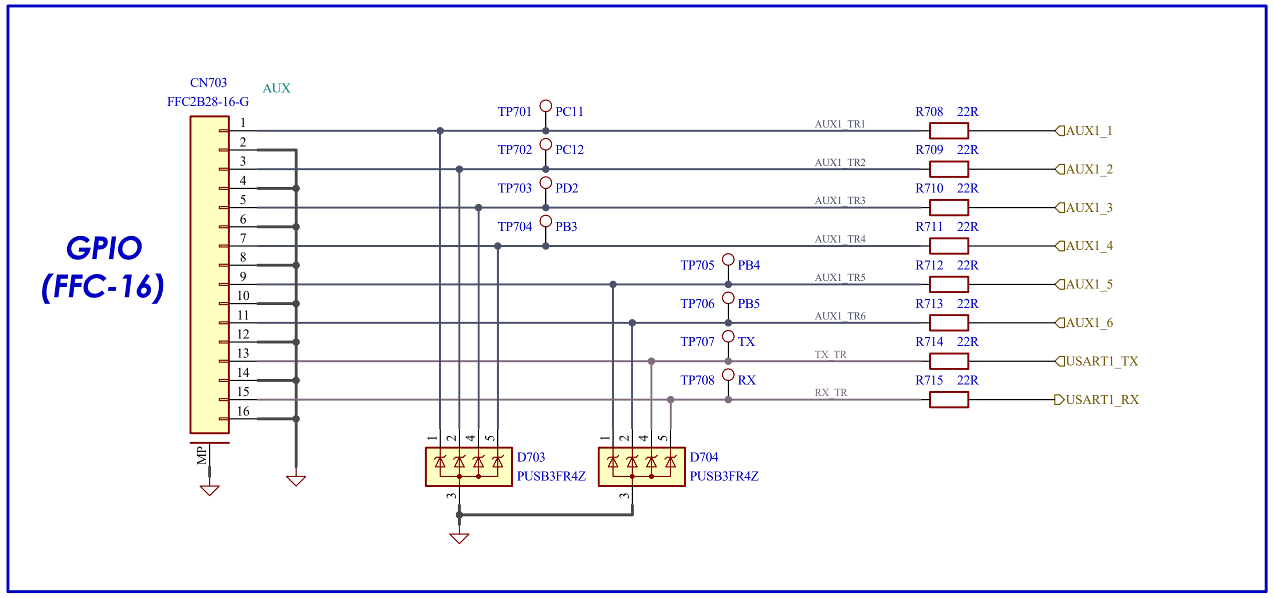
[7] Interface Interconnects

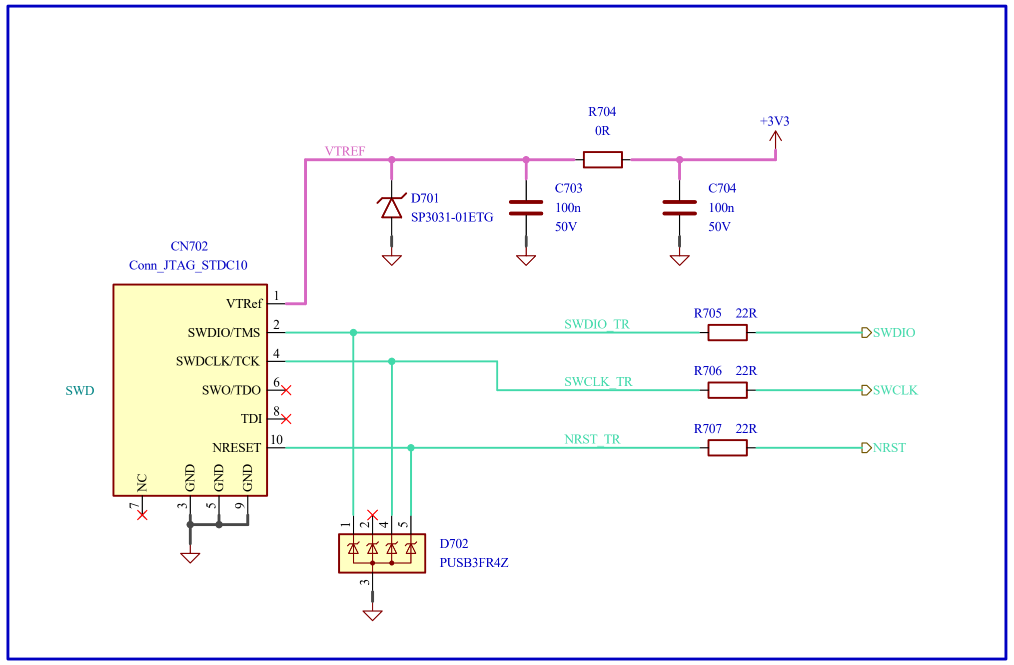
USB-C Connector



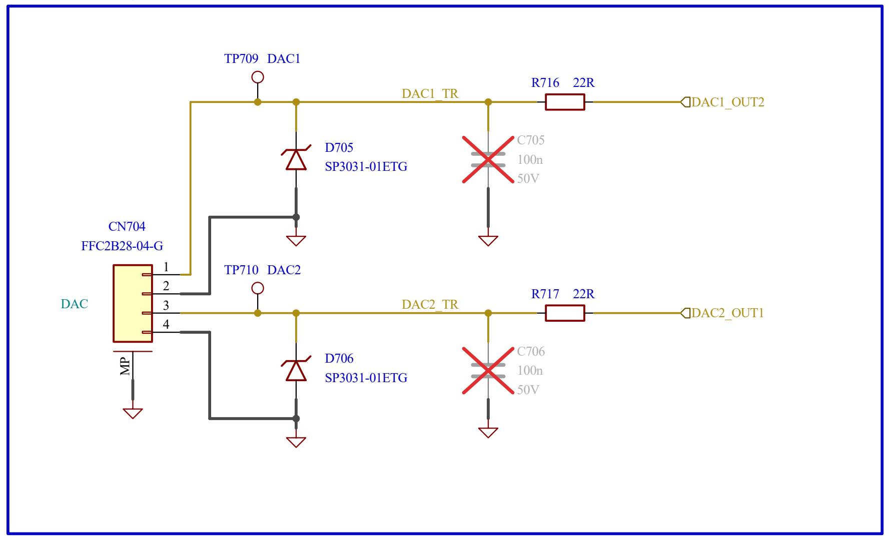
AUX Connector



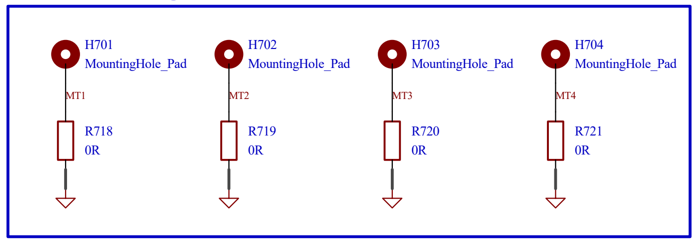
SWD Connector



Analog Output



Mounting Hole



Project: 2-Phase Interleaved Sync-Buck Converter		LQRx		
Title: **Interface Interconnects**	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:	
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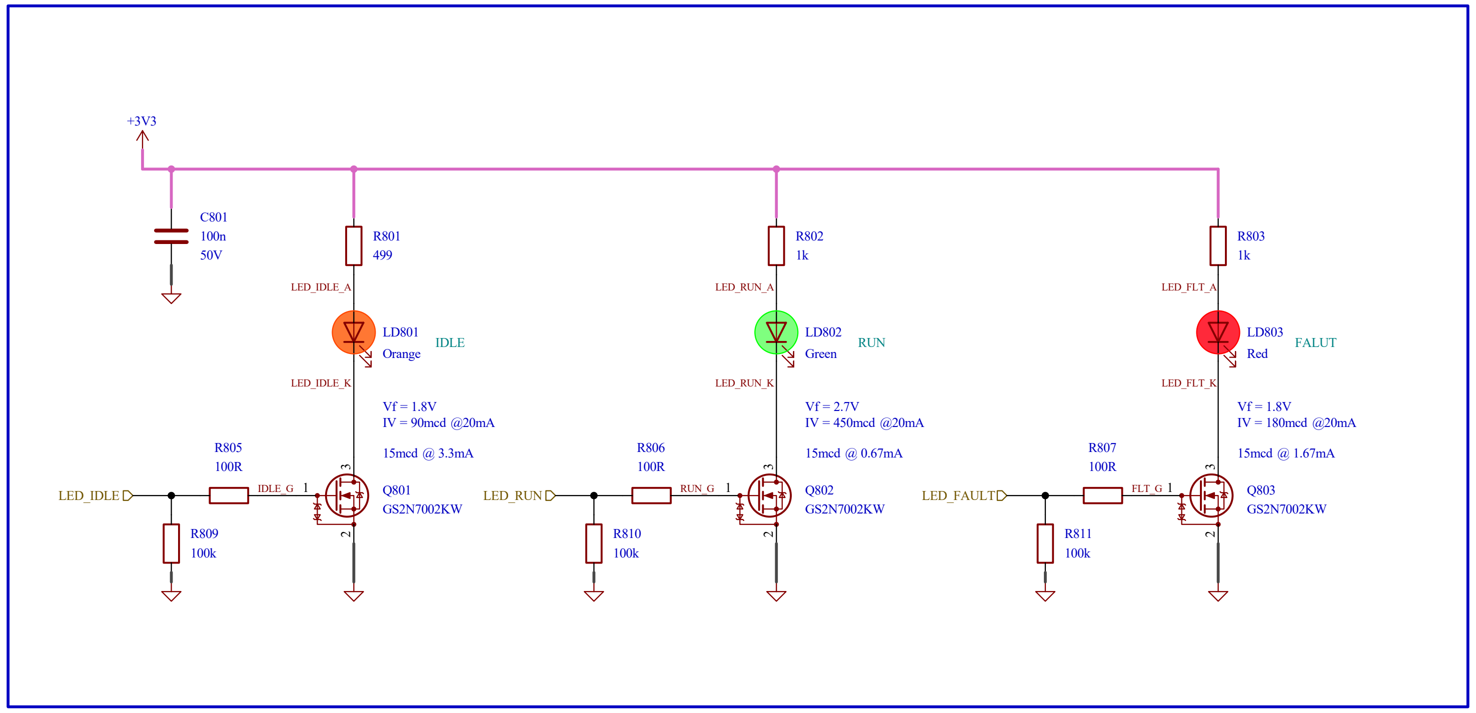
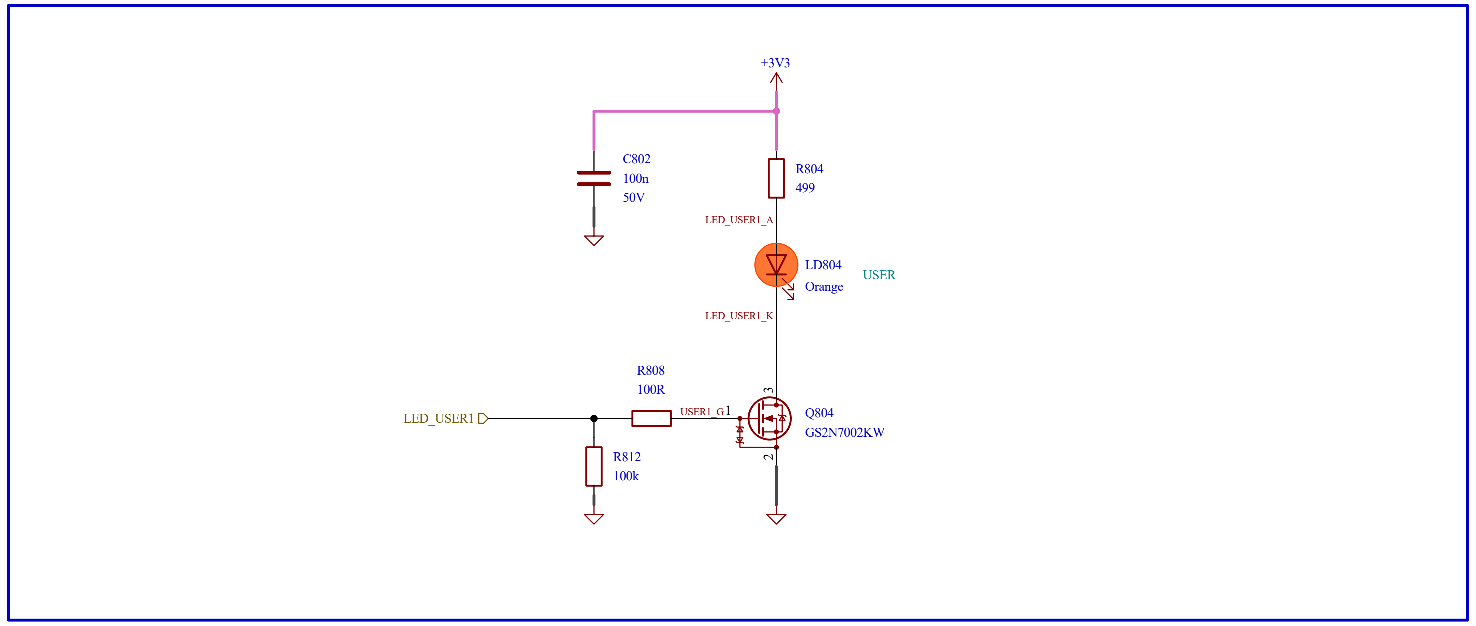
[8] User LEDs

Status LEDs



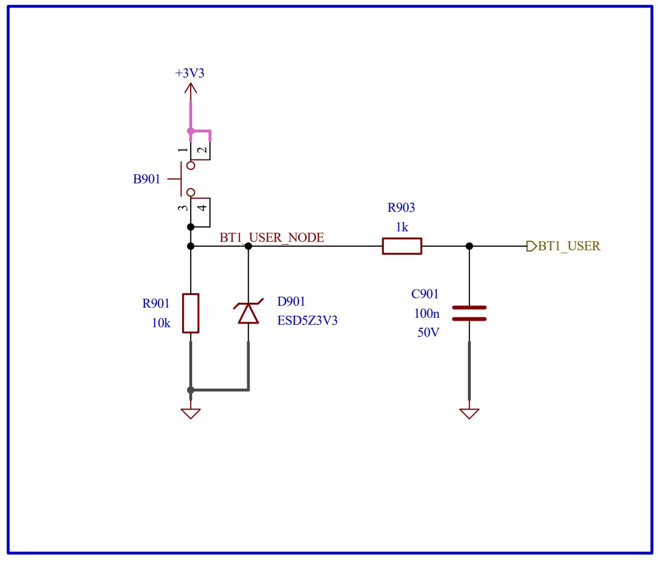
User LEDs



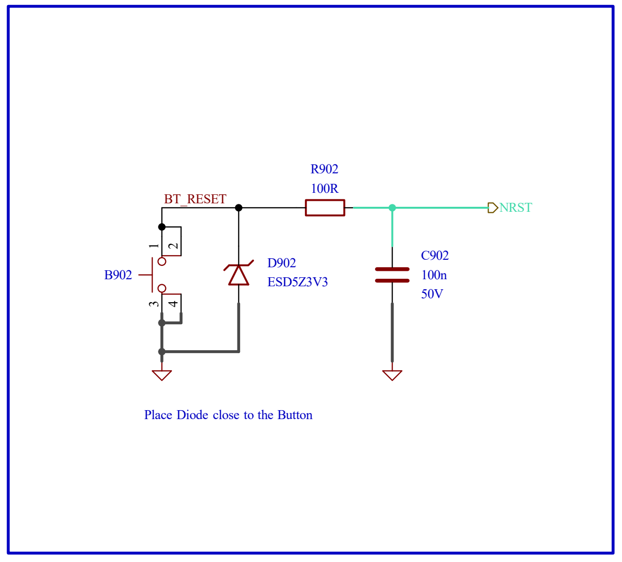
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx		
Title: User LEDs Status LEDs	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:	
	Reviewed by:	Date:	Rev: 1.0	Date: 03/06/2026
KiCad E.D.A. 10.0.3	Sheet: /Top/User LEDs/ File: user_LED.kicad_sch		Size A3	Sheet: 8 of 9

[9] User Interface

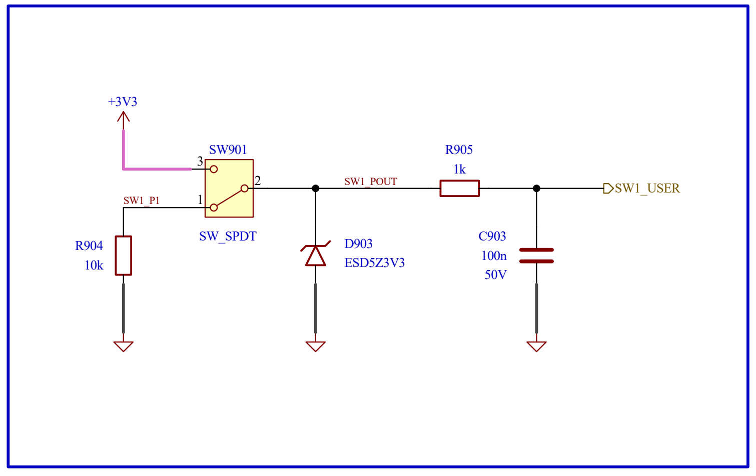
User Button



Reset Button



User Switch



Project: 2-Phase Interleaved Sync-Buck Converter		LQRx		
Title: User Interface Button & Switch	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:	
	Reviewed by:	Date:	Rev: 1.0	Date: 03/06/2026
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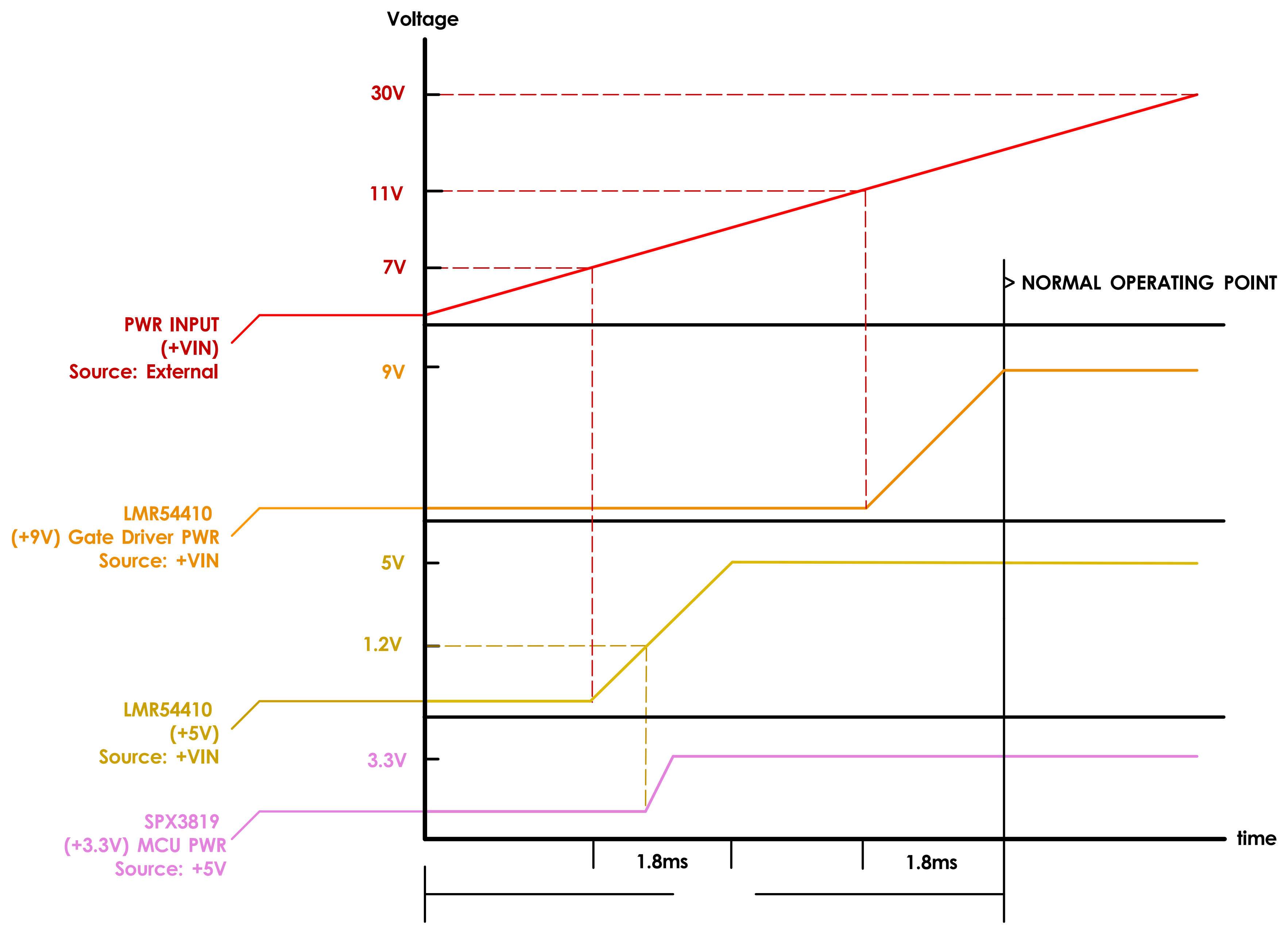
KiCad E.D.A. 10.0.3	File: user_interface.kicad_sch			

[D] Power Supply Sequencing



Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
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	Reviewed by:	Date:	Rev: **1.0** Date: 03/06/2026
KiCad E.D.A. 10.0.3	Sheet: /Power Supply Sequencing/ File: pwr_sequencing.kicad_sch		Size **A3** Sheet: **D** of **9**

[E] Revision History

Revison 1.0

- (18/05/2026) Initialize the project
- (03/06/2026) PC Board is Released

Project: 2-Phase Interleaved Sync-Buck Converter			LQRx		
Title: **Revision History**	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:		
	Reviewed by:	Date:	Rev: **1.0**	Date: 03/06/2026	
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KiCad E.D.A. 10.0.3	File: RevHistory.kicad_sch				